MATH 205

Linear Algebra

LECTURE 3 · COMPUTING A^{-1} & GAUSSIAN ELIMINATION

Why the method works, a full worked example, and a faster route that skips the inverse entirely.

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WELCOME BACK

Where Today Fits

Last time

- Uniqueness of the inverse, and $(AB)^{-1} = B^{-1}A^{-1}$
- The transpose, and symmetric / skew-symmetric matrices
- Elementary row operations and elementary matrices
- EA gives the same result as the row operation itself

Today

- Why $[A \mid I] \rightarrow [I \mid A^{-1}]$ actually works
- A full worked example, start to finish
- Gaussian elimination — a faster alternative

Course Roadmap

1. Systems of Linear Equations
2. **Matrices & Matrix Algebra**
3. Determinants
4. Vector Spaces
5. Eigenvalues & Eigenvectors

Solving Ax = b is fully in hand after today — determinants are next.

MOTIVATION

Where Systems Like This Come From

Real engineering problems rarely arrive as tidy 2-variable systems — they arrive like these.

Civil Engineering

Traffic flow around a town square: each intersection's cars-in must equal cars-out, giving one linear equation per intersection.

Chemical Engineering

Balancing a combustion reaction (ethane + oxygen → CO_2 + water) means matching atom counts — a linear system in the reaction coefficients.

Biology

A rabbit population's first-, second-, and third-year counts next year are linear combinations of this year's — the same system, year after year.

Astronomy

Fitting an asteroid's elliptical orbit to observed positions turns six data points into a linear system for the orbit's coefficients.

Computer Science

PageRank scores every web page's importance using its incoming links — Larry Page and Sergey Brin built Google on this linear system.

Any Engineering Field

Real models routinely involve dozens of equations in dozens of unknowns — solved by exactly the row reduction we're about to do.

The Chain of Elementary Matrices

Suppose $\theta_1, \theta_2, \ldots, \theta_n$ are the EROs that turn an invertible matrix A into I, with corresponding elementary matrices $E_1, E_2, \ldots, E_n$.

Applying them one after another to A:

$$\theta_n \cdots \theta_2 \theta_1 (A) = E_n \cdots E_2 E_1 A = I$$

Writing P for the product of all those elementary matrices:

$$PA = I, \qquad P = E_n \cdots E_2 E_1$$

This single equation ***PA = I*** *is the seed of everything that follows.*

Why This Gives Us A^{-1}

Since A is invertible, multiply both sides of PA = I on the right by A^{-1}:

$$PA = I \;\Rightarrow\; PAA^{-1} = IA^{-1} \;\Rightarrow\; PI = A^{-1}$$

PA = I means **A ~ I**. PI = A^{-1} means **I ~ A^{-1}**.

$$A \sim I \qquad \text{and} \qquad I \sim A^{-1}$$

The exact same row operations that turn A into I **also** turn I into A^{-1} — because they're the same operations, applied in the same order.

The Method

$$[\,A \mid I\,] \xrightarrow{\text{E.R.O.s}} [\,I \mid A^{-1}\,]$$

Along the way, some vocabulary we'll need:

— **Consistent:** Ax = b has at least one solution.

— otherwise **Inconsistent**: no solution at all.

Homogeneous

$$A\mathbf{x} = \mathbf{0}$$

always consistent — x = 0 is always a solution (the trivial solution).

Non-homogeneous

$$A\mathbf{x} = \mathbf{b}, \quad \mathbf{b} \neq \mathbf{0}$$

Solve by Finding A^{-1}

Solve this system by finding the inverse of the coefficient matrix:

$$\begin{aligned} 3x_1 + 4x_2 + 5x_3 &= 12 \\ x_1 - x_2 + 2x_3 &= 2 \\ 2x_1 + x_2 + 3x_3 &= 6 \end{aligned}$$

As a matrix equation AX = B:

$$\begin{bmatrix} 3 & 4 & 5 \\ 1 & -1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 2 \\ 6 \end{bmatrix}$$

Form [A | I] and row-reduce until the left half becomes I.

Row Reducing [A | I] — Steps 1–3

$$\left[\begin{array}{ccc|ccc} 3 & 4 & 5 & 1 & 0 & 0 \\ 1 & -1 & 2 & 0 & 1 & 0 \\ 2 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Start: [A | I]

→

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 1 & 0 \\ 3 & 4 & 5 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$R_1 \leftrightarrow R_2$$

Get a leading 1 in row 1

→

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 1 & 0 \\ 0 & 7 & -1 & 1 & -3 & 0 \\ 0 & 3 & -1 & 0 & -2 & 1 \end{array}\right]$$

$$\begin{aligned} R_2 &\to R_2 - 3R_1 \\ R_3 &\to R_3 - 2R_1 \end{aligned}$$

Clear below it in column 1

→

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & -2 \\ 0 & 3 & -1 & 0 & -2 & 1 \end{array}\right]$$

$$R_2 \to R_2 - 2R_3$$

Get a leading 1 in row 2

Continued on the next slide →

Row Reducing [A | I] — Steps 4–6

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 1 & 2 & -2 \\ 0 & 1 & 1 & 1 & 1 & -2 \\ 0 & 0 & -4 & -3 & -5 & 7 \end{array}\right]$$

$$R_1 \to R_1 + R_2$$
$$R_3 \to R_3 - 3R_2$$

Clear the rest of column 2

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 1 & 2 & -2 \\ 0 & 1 & 1 & 1 & 1 & -2 \\ 0 & 0 & 1 & \frac{3}{4} & \frac{5}{4} & -\frac{7}{4} \end{array}\right]$$

$$R_3 \to -\frac{1}{4} R_3$$

Get a leading 1 in row 3

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{4} & -\frac{7}{4} & \frac{13}{4} \\ 0 & 1 & 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & \frac{3}{4} & \frac{5}{4} & -\frac{7}{4} \end{array}\right] = [\, I \mid A^{-1} \,]$$

$$R_1 \to R_1 - 3R_3$$
$$R_2 \to R_2 - R_3$$

Clear the rest of column 3 — done

The Result

$$A^{-1} = \frac{1}{4}\begin{bmatrix} -5 & -7 & 13 \\ 1 & -1 & -1 \\ 3 & 5 & -7 \end{bmatrix}$$

Now solve X = $A^{-1}B$:

$$X = A^{-1}B = \frac{1}{4}\begin{bmatrix} -5 & -7 & 13 \\ 1 & -1 & -1 \\ 3 & 5 & -7 \end{bmatrix}\begin{bmatrix} 12 \\ 2 \\ 6 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x_1 = x_2 = x_3 = 1$$

One clean pass of row reduction now solves the system for any right-hand side B — that's the real payoff of finding A^{-1}.

A FASTER PATH

Gaussian Elimination

Finding all of A^{-1} is powerful — but if you only need to solve one system once, it's more work than necessary. Here's the direct route.

Row-Echelon Form

A matrix is in row-echelon form when:

1. The first nonzero number in any nonzero row is a 1 — a leading 1.
2. Rows that are entirely zero are grouped at the bottom of the matrix.
3. In any two successive nonzero rows, the leading 1 in the lower row sits farther right than the leading 1 above it.

Together, these three rules force a staircase pattern — each leading 1 sits down and to the right of the one before it.

Row-Echelon Form: Examples

$$\begin{bmatrix} 1 & 4 & 3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & 8 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

All three matrices above satisfy the definition — trace the staircase of leading 1's in each.

Note: this is echelon form, not fully reduced — entries above a leading 1 need not be zero (that stronger version is reduced row-echelon form, used by Gauss-Jordan elimination).

Same System, New Method

The exact system from before — this time we stop at echelon form, no augmented identity needed:

$$\left[\begin{array}{ccc|c} 3 & 4 & 5 & 12 \\ 1 & -1 & 2 & 2 \\ 2 & 1 & 3 & 6 \end{array}\right]$$

Start: [A | b]

→

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 3 & 4 & 5 & 12 \\ 2 & 1 & 3 & 6 \end{array}\right]$$

$$R_1 \leftrightarrow R_2$$

Leading 1 in row 1

→

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 7 & -1 & 6 \\ 0 & 3 & -1 & 2 \end{array}\right]$$

$$R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - 2R_1$$

Clear below it

→

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & -1 & 2 \end{array}\right]$$

$$R_2 \to R_2 - 2R_3$$

Leading 1 in row 2

Half the columns of the inverse method — since there's no identity block to carry along.

Finishing with Back-Substitution

Two more steps reach echelon form:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -4 & -4 \end{array}\right]$$

$$R_3 \to R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$R_3 \to -\tfrac{1}{4}R_3$$

Echelon form gives the system directly, no augmented identity to read off:

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 2 \\ x_2 + x_3 &= 2 \\ x_3 &= 1 \end{aligned}$$

Back-substitute from the bottom row up:

$$x_3 = 1 \;\Rightarrow\; x_2 = 2 - x_3 = 1 \;\Rightarrow\; x_1 = 2 + x_2 - 2x_3 = 1$$

Two Practical Notes

1

Don't undo progress

In the row reduction process, don't perform steps by which you lose zeros or 1's you've already obtained. Each step should move you forward, never backward.

2

Avoid fractions when you can

If a different valid row operation keeps entries as whole numbers, prefer it — fractions are correct but easy to slip up on by hand.

TRY IT YOURSELF

Practice 1 — Row Reduction Drill

Row reduce this matrix all the way to row-echelon form:

$$\begin{bmatrix} 0 & -7 & -4 & 2 \\ 2 & 4 & 6 & 12 \\ 3 & 1 & -1 & -2 \end{bmatrix}$$

Hints

- No row currently starts with 1 — swap or scale to get one
- Clear every other entry in that column before moving on
- Keep going until every column with a pivot is completely clean

Source: Margalit & Rabinoff, Interactive Linear Algebra, Section 1.2 — pure algorithm practice, no story problem attached.

TRY IT YOURSELF

Practice 2 — Solve via Row Reduction

Solve this system using row reduction:

$$\begin{aligned} 2x + y + 12z &= 1 \\ x + 2y + 9z &= -1 \end{aligned}$$

Hints

- Only 2 equations, 3 unknowns — expect a free variable
- Reduce to echelon form, then write x and y in terms of z

Source: Margalit & Rabinoff, Interactive Linear Algebra, Section 1.2.

RECAP

Key Takeaways

Why the method works

the EROs that turn A into I are the same ones, in the same order, that turn I into A^{-1}.

The recipe

[A | I] row-reduces to [I | A^{-1}] — read the inverse straight off the right half.

Solving with A^{-1}

once you have it, $X = A^{-1}B$ solves the system for any right-hand side B.

Row-echelon form

leading 1's staircase down and to the right; zero rows sink to the bottom.

Gaussian elimination

reduce to echelon form, then back-substitute — faster when you just need one solution.

Two methods, same answer

the inverse method and Gaussian elimination always agree — pick based on what you need.

NEXT TIME

Looking Ahead

Lecture 4

Gauss-Jordan & the Big Picture

- Gauss-Jordan elimination: reducing all the way to RREF
- No back-substitution needed — read off the answer directly
- The Invertible Matrix Theorem: one master result
- Many equivalent ways to say “A is invertible”

$$A \text{ invertible} \iff A\mathbf{x} = \mathbf{b} \text{ has a unique solution for every } \mathbf{b} \iff \operatorname{rref}(A) = I$$

Homework 2

Reference: Anton & Rorres, Sections 1.4 & 1.5

Questions?

Computing A^{-1} & Gaussian Elimination · Lecture 3 complete

NEXT: GAUSS-JORDAN & THE BIG PICTURE